

White Abalone Conservation in the Santa Barbara Channel

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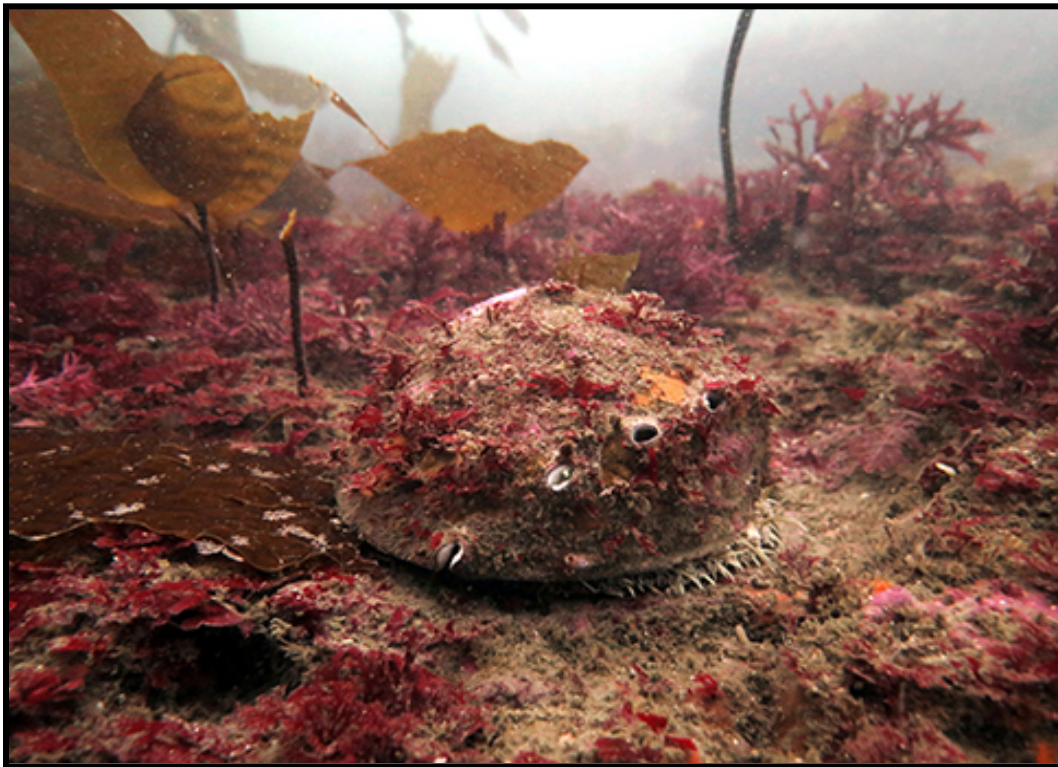


Figure 1. Rare White Abalone found off S. CA Coast (A. Maguire)

Executive Summary

This conservation plan focuses on restoring populations of the endangered white abalone (*Haliotis sorenseni*) in the kelp forest ecosystems of the Santa Barbara Channel. Once abundant along the Southern California coast, white abalone populations have declined by approximately 99% due to historical overharvesting, habitat degradation, and environmental stressors. Because white abalone reproduce through broadcast spawning, extremely low population densities have reduced successful reproduction and genetic diversity, placing the species at risk of extinction. This plan aims to address these challenges by restoring healthy kelp forest habitats and increasing the density and resilience of white abalone populations.

The project centers on kelp forests within the Santa Barbara Channel, a region extending from Point Conception to Point Dume and encompassing the Northern Channel Islands. Healthy kelp forests provide essential habitat and food sources for white abalone and support a diverse marine ecosystem. However, these ecosystems are increasingly threatened by marine heat waves, warming waters, hyperpredation, and overgrazing by sea urchins. By improving kelp forest conditions and reducing ecological stressors, this plan seeks to create a more suitable environment for the recovery and long-term sustainability of white abalone.

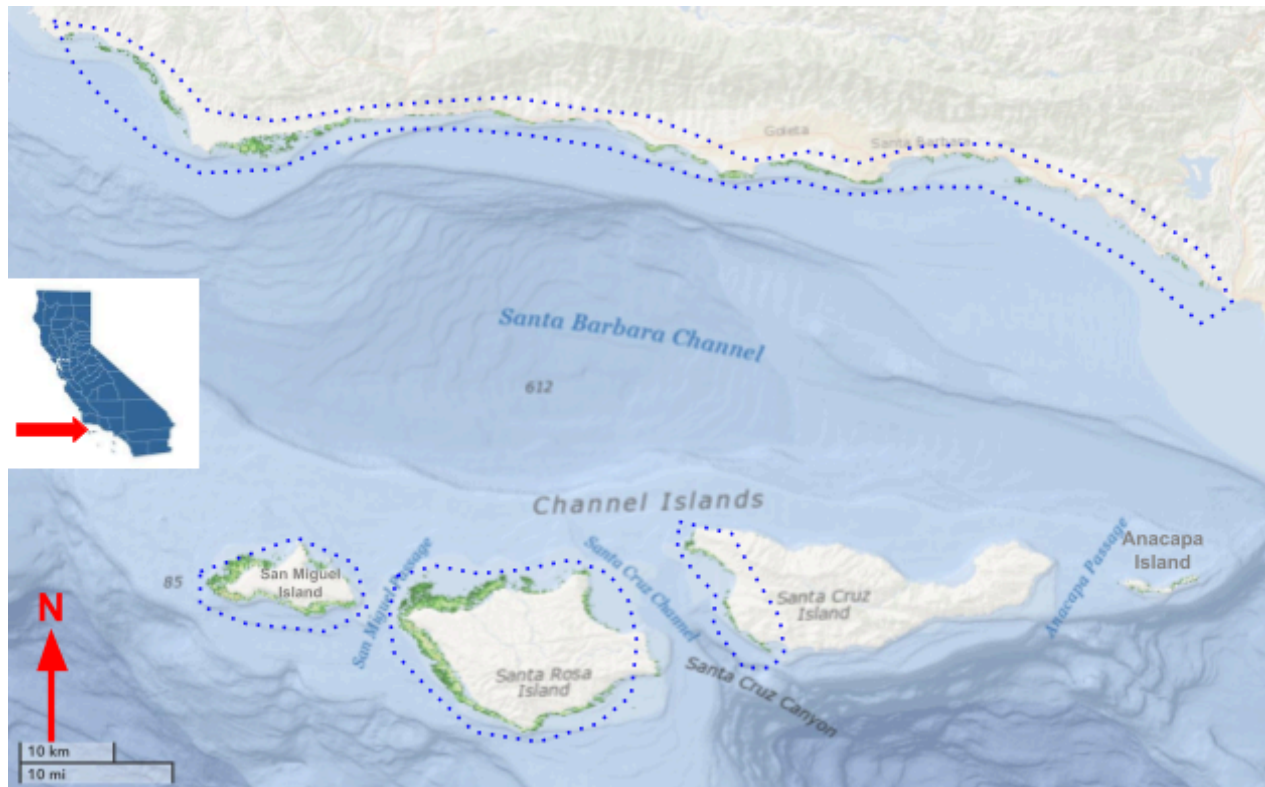
The conservation strategy emphasizes collaboration among scientists, government agencies, Indigenous communities, and conservation organizations. Key partners include NOAA, the Channel Islands National Marine Sanctuary, research institutions, aquaculture facilities involved in captive breeding, and the Chumash people, whose traditional ecological knowledge and cultural relationship with abalone provide valuable guidance for restoration efforts.

Primary objectives include increasing white abalone population density, restoring kelp forest habitat, reducing sea urchin overgrazing, and supporting successful reproduction through captive breeding and outplanting programs. Additional goals include strengthening ecosystem resilience, supporting cultural connections to abalone, and promoting long-term ecological monitoring and management. Our goal is to increase the coverage and density of degraded kelp forests by 30-40% over the next 50 years. Also, we hope to have a healthy population of white abalone in 5-10 sites within a minimum of 50 years. In terms of social goals, by 2030, increase local community and stakeholder engagement by 30% and form at least two partnerships with Chumash organizations. Over the next 50 years, our goal is to increase economic profit by 30% from restored kelp forest habitat (Eger et al.). Within a minimum of 50 years, White Abalone populations should be able to reproduce successfully without captive breeding programs. Through coordinated restoration, scientific research, and community engagement, this plan aims to support the recovery of white abalone while enhancing the health and biodiversity of the Santa Barbara Channel's marine ecosystems.

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Project Scope



*Figure 2. Map of Persistent Kelp Forests Locations in the Santa Barbara Channel, California
Map created using data from the CDFW Marine Biogeographic Information System*

The Santa Barbara Channel is the northernmost part of the Southern California Bight. It is a 130 km-long marine passage extending from Point Conception in Santa Barbara County to Point Dume in Ventura County and extending out to the four Northern Channel Islands (Seagrant, n.d.). Our scope encompasses the kelp forests within the Santa Barbara Channel, which are home to our focal species, the white abalone. The degradation of kelp forest habitat along the west coast led to a 99% decrease in white abalone populations (NOAA, 2025). At such low densities, remaining white abalone populations are incredibly sparse. As broadcast spawners, effective reproduction requires higher densities, leading to a collapse in breeding success that has led to reduced genetic diversity, leaving remaining populations more vulnerable to stressors and disease, particularly withering syndrome. (Xerces Society, n.d.). Working to improve the health of these kelp forests by increasing density will have positive cascading effects on the restoration efforts for the white abalone.

Project Vision

The vision of this project is to restore the health, biodiversity, and abundance of the Santa Barbara Channel's kelp forests by reducing the adverse impacts of marine heat waves, hyperpredation, overgrazing by sea urchins, and warming waters in order to support resilient white abalone populations (Eisaguirre, 2020). These efforts will benefit the Channel's broader marine ecosystem, help restore abalone population, genetic diversity, and successful reproduction, support the cultural appreciation and use of abalone by the Chumash people, aid other abalone species and a wide array of other marine organisms, and strengthen coastal communities overall.

Planning Context

White abalone (*Haliotis sorenseni*) are an incredibly important native species that help maintain the biodiversity of the Santa Barbara Channel kelp forests. The white abalone are ecosystem architects that increase biodiversity by feeding on kelp, clearing space for multiple kelp species that draw in an array of fish and other organisms. It has held cultural significance and had practical uses for thousands of years (NOAA, 2023). Abalone is integrated into the spiritual practices of the Chumash and Yokut in their burial rituals, while also being used for tools and jewelry ("The cultural", 2023). This plan seeks to address the main threats brought to kelp forests due to their critical role in the survival of the white abalone after captive release. Some of the main points we are concerned with are reducing urchin populations and improving the density of the kelp forest while also making it more resistant to warming waters. These efforts will create an improved habitat for the white abalone, facilitating an easier recovery and allowing for denser, healthier populations.

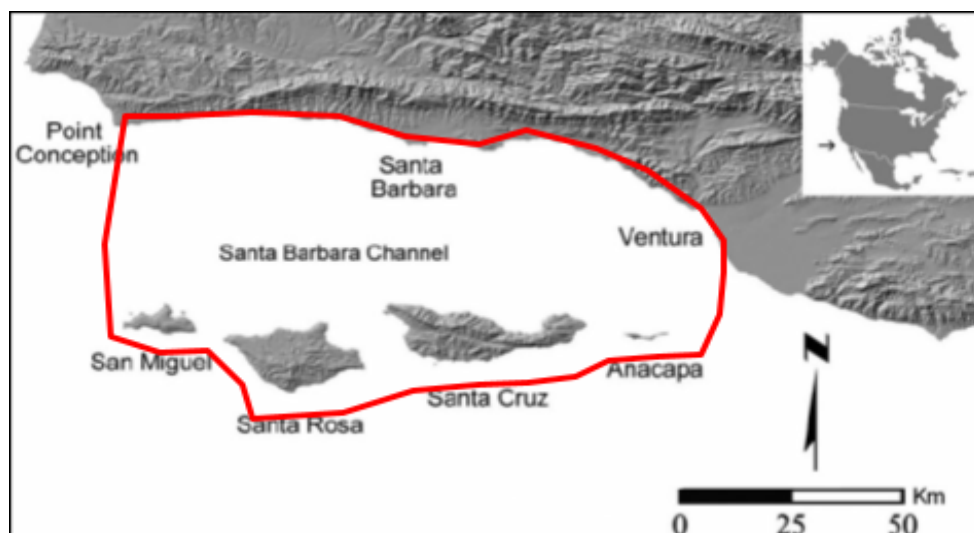


Figure 3. Map of the Santa Barbara Channel and Focus of Conservation Area (Erlandson, J. (2009))



Figure 4. White Abalone (NOAA Fisheries, 2009))



Figure 5. Kelp Forest Floor (Bob Miller, 2021)

Planning Team and Management Process

For this conservation plan, our team wanted to work, specifically, with the Chumash People. We would like to have their input on where, geographically, we should focus our project, as well as on sustainable and ethical ways to proceed. We would like to highlight their addition of traditional ecological knowledge (TEK) in the project to further educate individuals on how knowledgeable all indigenous tribes are. NOAA (National Oceanic and Atmospheric Association) is another partner we would like to have a stake in this project. With the multitude of knowledgeable data scientists and conservationists who work for NOAA, the organization can contribute to data collection while also pointing the project in the right direction. The Office of Habitat Conservation, which is within NOAA, has a culmination of data scientists, biologists, and conservation ecologists who can hopefully provide the team with helpful insight and previous data collections as well (Office of Habitat Conservation, n.d.). As the scope of our

project lies within federal waters, we would also like to include the Channel Islands National Marine Sanctuary (CINMS) to gain more of the federal government's attention on this issue.

Table 1 - Planning Team

Member	Skillset	Organization	Role
Michael Torbensen	Collaboration and communication liaison	UCSB Environmental Studies Department	Community Liaison
Avery Morgan	White Abalone Expert	UCSB Environmental Studies Department	Researcher on the role of White abalone in marine ecosystems
Kate Campagne	White Abalone population and breeding specialist	UCSB Environmental Studies Department	Researcher on Abalone Reproduction and Population Dynamics
Piers Nicklin	Conservationist, project management, marine species monitoring	UCSB Environmental Studies Department	Restoration and Conservation Coordinator, Legal Liaison
Lauren Combest	Data and research collection, science communication, and knowledge of temperature effects	UCSB Environmental Studies Department	Researcher on kelp outplanting and temperature resistance

Shareholders

Key partners and shareholders that make up our community include government agencies, the indigenous Chumash tribe, non-profits, scientific and regulatory agencies, and research labs. Together, this community has the power to aid in our White Abalone conservation efforts through the conduct of monitoring, scientific dives, implementing and enforcing policies and laws, breeding and outplanting abalone and kelp, engaging the public, incorporating traditional ecological knowledge, and supporting a long-term recovery of White abalone and surrounding marine ecosystems.

Table 2 - Shareholders

Member	Organization	Role
Dr. Kristin Aquilino	CA Dep. of Fish & Wildlife (California Department of Fish and Wildlife, 2017)	Issuing permits and licenses needed; CDFW employees manage and implement programs
Dr. Dawn Murray, Dr. Dan Rogers-Bennett Dr. Melissa Neuman	NOAA; NOAA Fisheries (What is NOAA, n.d.)	Leading in white abalone conservation, recovery plans, captive breeding and recovery, monitoring populations, enforcement, and public outreach, Financial support
Chris Mobley	Channel Islands National Marine Sanctuary Foundation (Channel Islands National Marine Sanctuary, n.d)	Restoration, public engagement, outplanting captive-bred abalone,
Chumash Tribe Member & Elder	Chumash Tribal Council (Van Hoorn, 2022)	Long-term ecological reliance on abalone, cultural advisors/shareholders, and TEK of coastal ecosystems from the Coastal Band, Barbareño Band, and Santa Ynez Band
Jenn Caselle	Jenn Caselle Lab (Jenn Caselle Lab, n.d.)	Academic research, regular monitoring of channel islands; scientific divers can physically implement any parts of the conservation plan that require the addition or removal of species
Carol Blanchette	Pisco Lab (Partnership for Interdisciplinary Studies of Coastal Oceans, n.d.)	Long-term research + monitoring of aquatic activity within our scope; share data sets on similar aquatic ecosystems; scientific divers can physically implement any parts of the conservation plan that require addition or removal of species
Devin Spencer	Cultured Abalone Farm (The Cultured Abalone Farm, n.d.)	Captive breeding of red and green abalone, white abalone, and can focus on white abalone more through government subsidies or grants.

Sideboards

The goal is to restore the white abalone to a healthy, self-sustaining population. There are a number of legal restrictions that can complicate this process, one being that the white abalone is listed as an endangered species, and this means permits would need to be obtained from the U.S. Fish and Wildlife Service before conservation efforts can begin. The permitting process will take time and needs to be considered in the overall timeline of our conservation plan (*U.S. Fish & Wildlife Service*). Another issue is that the desired location for the implementation of our kelp forest restoration falls under multiple jurisdictions. San Miguel, Santa Barbara, and Anacapa, as well as parts of Santa Cruz and Santa Rosa, are owned by the National Park Service. While the Nature Conservancy owns most of Santa Cruz, the U.S. Navy owns San Nicolas and San Clemente. Finally, Santa Catalina is owned by Catalina Island Conservancy and private interests. Collaboration with the separate groups that own the islands is crucial in making sure our efforts will be effective in restoring white abalone populations.

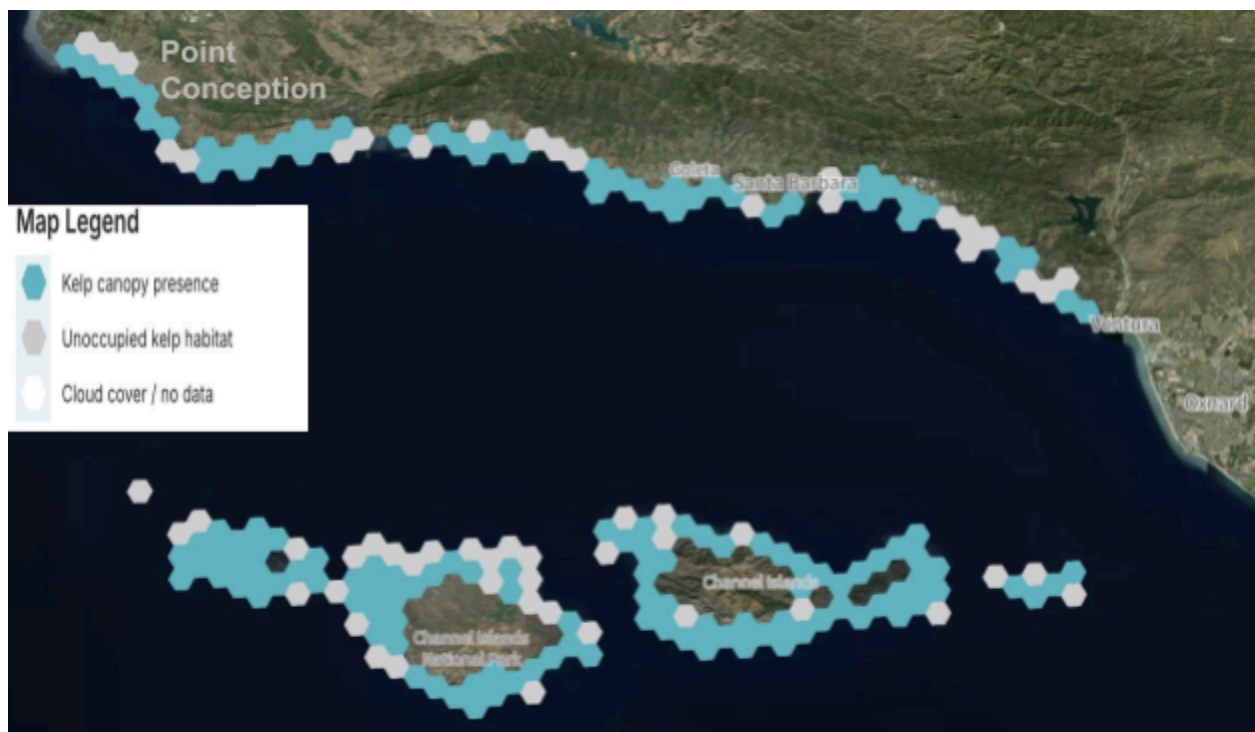


Figure 6. Implemented Kelp Restoration Areas (Kelp Watch 2025)

Constraints

Conservation efforts toward white abalone in the Santa Barbara Channel are influenced by constraints. These constraints fall under several categories:

- Physical:
 - Field Monitoring and conservation implementation require scuba diving, often deep, anywhere from 55-130ft+
 - Fair Weather and low swell to ensure safe boat travel and safe diving
 - Limited to rocky reefs where white abalone and giant kelp attach to
 - Marine heatwaves prove detrimental to both the survival of white abalone and giant kelp
- Legal:
 - White abalone fall under the Endangered Species Act, requiring special permitting to raise in captivity
 - Permitting from the CA Fish and Wildlife to conduct research on a marine species
- Economic:
 - Fuel for boat travel
 - Costs of raising and maintaining captive-bred white abalone
 - Research to find and monitor existing wild white abalone populations
 - Costs of equipment for diving, breeding, and research
- Social:
 - In non-MPA restoration sites, conflicting interests with fishing practices or recreational use could impact the long-term success of the project
- Technical:
 - Deepwater diving requires experienced divers who are trained and certified to perform research.
 - Captive breeding of white abalone takes time to raise the abalone till they are around 1 year old, as well as space capacity, limiting the amount of abalone that can be raised at once
 - Long-term monitoring equipment needs to be maintained or replaced
 - Locating white abalone can be one of the most difficult challenges, with sparse populations. The use of eDNA sampling of the water can include mucus, feces, or tissue from white abalone.

Fundamental Objectives and Desired Outcomes

The fundamental objectives table (*Table 3*) serves to provide a clear outline for the conservation plan, highlighting fundamental objectives, features, indicators, and goals that ultimately align with shareholder values and long-term restoration goals.

Table 3 - Fundamental Objectives

	Fundamental Objectives	Features	Indicators	Goals
	<i>What & Direction</i>	<i>What can be measured</i>	<i>How will 'what' be measured</i>	<i>Goal/time</i>
Environmental – Biodiversity	- Reduce purple urchin populations in urchin barrens and around kelp forests	- The population density of purple urchins in affected areas can be observed	- Divers perform subtidal belt transects within fixed quadrats	- Keep purple urchin populations below the threshold of urchin barrens every year
	- Increase kelp forest density and coverage in the SB Channel	- Comparisons to what was previously observed through the % change.	- Satellite observed kelp cover, - Density per 1mi ²	- Increase the coverage and density of degraded kelp forests by 30-40% over the next 50 years
	- Increase White Abalone populations (National Marine Fisheries Service, 2008)	- Density and abundance monitoring programs every 5 years - size frequency	- Partner with NOAA and other NGOs to create monitoring programs. Use of subtidal belt transects, quadrats, and ROVs	- A healthy and self-sustaining population in 5-10 sites within a minimum of 50 years
Environmental – Ecological Processes	- Restore natural reproduction rates in white abalone within and around CA giant kelp forests	- Reproduction success in White Abalone, measuring abalone population density	- Monitoring of White Abalone populations with transect sampling methods	- Within a minimum of 50 years, White Abalone populations should be able to reproduce successfully without captive breeding programs
Environmental –Ecosystem Services	- Restore their role as habitat engineers working to graze algae cover	- Algal % cover relative to giant kelp cover	- Divers perform subtidal belt transects, using randomly placed quadrats	-Show increased biodiversity relative to the 2026 baseline, over the next 50 years, provide self-sustaining grazing from white abalone

Economic	- Resilient ecosystems that contribute to fisheries	- Stock assessments - Bycatch and habitat impact assessments - Economic value of restored habitat	- Fishery observers/ at sea monitors - Revenue from fisheries, ecotourism	- Over the next 50 years, increase economic profit by 30% from restored kelp forest habitat (Eger, A.et al)
Social	- Restore indigenous culture/tradition	- Indigenous Chumash levels of abalone use - Number of tribal partnerships	- Interviews with Chumash tribe elders - Tracking number of official partnerships - Monitoring participation from Chumash in restoration and education.	- By 2030, form at least two partnerships with Chumash organizations
	- Increase in recreation, eco-tourism, and environmental education	- Number of tourists - Education program participants - Public awareness	- Visitor count from parks, beaches, tourism companies - Visitor count from education programs - Public surveys	- By 2030, increase local community and stakeholder engagement by 30%

Environmental Biodiversity:

In order to increase environmental biodiversity in this conservation plan, we plan to focus on three fundamental objectives. The first being to **increase white abalone populations** in both their current and historic ranges. As broadcast spawners, population density is critical to reproductive success and the main reason for their extremely low population size, which is why this is our biggest objective in this conservation plan. To reintroduce white abalone to their historic ranges, we must first look at the stability of the ecosystem to which we are releasing them. To ensure the stability of the kelp forest ecosystem, or in many cases along the channel, completely restore the area. Urchin barrens are degraded rocky reef habitats, covered with dormant purple urchins(*Strongylocentrotus purpuratus*) and devoid of kelp due to over-grazing by the urchins. To reverse these barrens, we plan to **reduce purple urchin populations**, allowing the reef to regrow giant kelp(*Macrocystis pyrifera*), aligning with our final objective to restore biodiversity, which is to **increase kelp forest cover**. Increasing kelp forest cover will allow for a stable ecosystem where biodiversity can return, including the white abalone.

Ecological Processes:

Restoring natural reproduction rates of white abalone would both allow them to have a self-sustaining population in the wild and increase nutrient cycling within and around California giant kelp forests. Reproductive success in white abalone depends on high population density, and without it leads to a collapse of breeding success and, in many cases, complete loss of white abalone in certain regions. Nutrient cycling is a critical aspect in kelp forests, and white

abalone play a key role. By converting algae into dissolved nutrients, supporting primary productivity.

Ecosystem Services:

Our fundamental objective for ecosystem services is to **restore the white abalone's role as habitat engineers for kelp forest ecosystems**. Working to graze algae cover, white abalone play a key role in regulating algal overgrowth, maintaining biodiversity, and overall ecosystem stability. Without the presence of white abalone, faster-growing algal turf outcompetes giant kelp and dominates the rocky reef. Restoring their role as habitat engineers will help the long-term resilience of the ecosystem.

Economic:

Ensuring long-term resilience in ecosystems is critical in maintaining sustainable fisheries practice. Healthy, stable kelp forest ecosystems serve as critical habitats and are essential for reproductive success among many of the fish that support fisheries and feed communities. When kelp forests are degraded as a result of algal overgrowth and urchin barrens, the result is an overall loss in biodiversity and ecosystem productivity. Maintaining the resilience of these ecosystems provides economic benefit and value.

Social:

Our fundamental social objectives are to restore **indigenous Chumash cultures and traditions**. Historically, the Chumash have relied on abalone and the kelp forest ecosystem, forming strong cultural ties. Today, they are leading in marine conservation; forming MPAs that prohibit development, manage sustainable fishing, and protect numerous species. We also aim to **increase recreation, eco-tourism, and environmental education**. By doing this, we increase the awareness of the ecological importance of kelp forest ecosystems and the species that depend on them, like the white abalone. Eco-tourism and increased recreational activity can also be profitable, supporting the research and conservation of these areas. Being open in the process of planning and implementing conservation strategies allows for increased public support and understanding, something that is essential for the long-term preservation of these habitats and species.

Conservation Features

Features, indicators, and targets:

Biodiversity:

Our fundamental objectives to increase biodiversity are to reduce purple urchin populations, increase kelp forest density and coverage, and to increase the white abalone

population in the Santa Barbara Channel. In order to achieve these objectives, we need something measurable, how it can be measured, as well as a tangible, quantifiable goal. To reduce purple urchin populations, we plan to directly remove them and to track our effect by measuring the population density of purple urchins in affected areas. To effectively measure population density, divers will perform subtidal belt transects and randomly placed quadrats. Our goal is to keep urchin populations below the threshold of becoming an urchin barren, repeating the process annually. To track the increase in kelp forest density and coverage, we will monitor kelp coverage and calculate the percent change from current images to a 2026 baseline. We will do this through satellite observations of kelp cover and noting the density per square mile. Our end goal is to increase the coverage and density of degraded kelp forests by 30-40% over the next 50 years. If that number seems high, it is only half of the kelp forest coverage lost off the California coast over the past 50 years. To track the effect of increasing white abalone populations, we will measure the density and abundance of our restoration sites every year. We will accomplish this through the creation of monitoring programs, utilizing subtidal belt transects, ROVs, and eDNA testing. Our goal is to have a healthy population of white abalone in 5-10 sites within a minimum of 50 years.

Ecological Processes:

Our fundamental objective toward ecological processes is to restore natural reproduction rates in white abalone. To feature this, we will measure the population density and the number of adolescent white abalone. We will do this through long-term monitoring of white abalone populations, using transect methods. Our goal is to restore successful reproduction rates in White abalone within a minimum of 50 years and without captive breeding. In accomplishing this, we also increase nutrient cycling, supporting ecosystem productivity.

Ecosystem Services:

Our fundamental objective is to restore a key service that white abalone provide to their ecosystem. Their role as habitat engineers is critical in maintaining ecosystem biodiversity by managing algal overgrowth. To track the effectiveness of our efforts, we will measure the algal percent cover in relation to giant kelp cover. In order to accomplish this, divers will perform subtidal belt transects and use randomly placed quadrats, calculating percent cover among different algal species. Our goal is to restore the white abalone's critical role, showing increased biodiversity relative to our 2026 baseline, over the next 50 years.

Economic:

Our economic fundamental objective is to create resilient ecosystems that contribute to fisheries. We will measure our effect by looking at stock assessments, bycatch, habitat impact assessments, and the economic value of the restored habitat. We will do this by utilizing fishery observers and monitoring the revenue from fisheries. Our goal is to, over the next 50 years, increase the economic profit by 30% as a result of restored kelp forest habitats. By doing this, we

can increase profit significantly, up to \$147,100 per hectare, through increased fisheries profits, nitrogen fixing, and carbon sequestration.

Social:

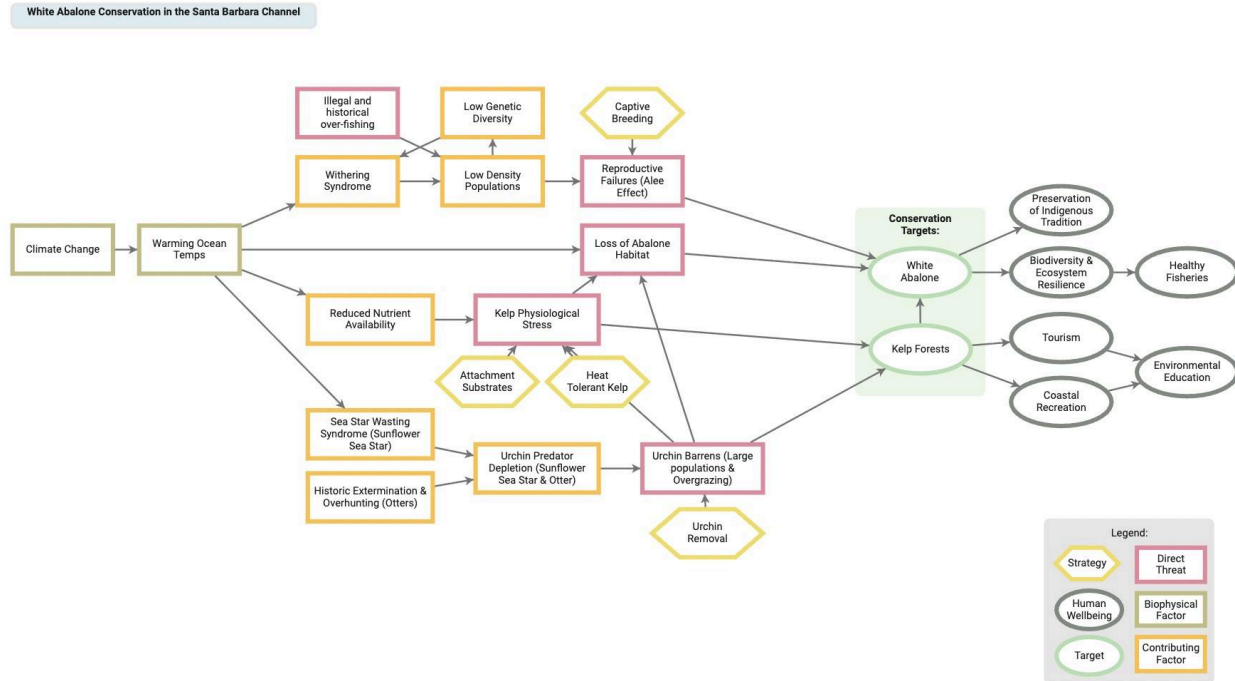
Our fundamental objectives are to restore indigenous cultures and traditions and to increase recreational use, eco-tourism, and environmental education. To track our progress, we will measure the level of Chumash abalone use and participation in conservation efforts, as well as the number of our tribal partnerships. We will do this through interviews with Chumash elders, tracking the number of official partnerships, and monitoring the participation of Chumash in restoration and education. Our goal is to form at least two partnerships with Chumash environmental organizations by 2030. Our measurable features to track an increase in recreation, eco-tourism, and environmental education are to monitor the number of tourists, participants in education programs, and the amount of public awareness. We will do this by tracking the visitor count from parks, beaches, tourism companies, and educational programs, as well as public surveys. Our goal is to by 2030 increase local community and stakeholder engagement by 30%.

Situation Analysis

Miradi's situation analysis model, shown in Figure 7, is a tool used by conservation planners that provides visual diagrams to map out the project scope, identify conservation targets, and analyze the contributing factors that impose direct and indirect threats to them. This representation helps understand relationships within the biological environment as well as social, cultural, economic, and political systems that may affect our focal values (Conservation Measure Partnership, 2025).

Conservation targets (green oval) represent the target species and ecosystem intended to be protected and improved. Human well-being targets (grey oval) represent components of human life that are affected by the status of conservation targets. Contributing factors (orange rectangle) include indirect threats, opportunities, or other factors that lead to indirect threats. Direct threats (red rectangle) are immediate processes that cause the decline of targets, while contributing factors (orange rectangle) indirectly intensify these direct threats. Strategies (yellow hexagon) are imposed plans to mitigate threats and improve the overall health of targets.

Figure 7. Situation Analysis



The primary environmental target for this analysis is the white abalone (*Haliotis sorenseni*), while the human well-being targets include the preservation of indigenous traditions and biodiversity/ecosystem resilience, which in turn provide healthy fisheries. The most consequential direct threat to the white abalone is reproductive failure (ie, Alee Effect). The Alee Effect is a result of abalone populations decreasing to such low densities that eggs from females and sperm from males rarely meet in the water column (Stierhoff et al., 2012). Reproduction events themselves are much more uncommon for the white abalone than for other abalone species. Sometimes, they can initiate their broadcast spawn fewer than once per 8 years. The lower the white abalone population densities get, the harder it is for them to successfully reproduce. This is where captive breeding strategies can shine. If successful and many juvenile abalone survive long enough to attempt reproduction strategies, the Alee Effect may be significantly reduced.

White abalone population densities have also been directly threatened by historic overfishing and illegal fishing. In 1955, this species was listed as “a species to harvest” by the CDFW. These abalone, along with all other abalone, continued to be harvested by the thousands until 1996, when the declining population numbers resulted in a closure of the fishery (DiNardo et al., 2021). The closure was a very positive action taken by California that nearly saved the species. As the numbers continued to dwindle, the National Marine Fisheries Service concluded that the white abalone was endangered under the Endangered Species Act. Despite efforts, the damage caused by the fishing industry is irreversible.

California's Central Coast, including the northern and southern coastlines, has seen a rapid decline, as much as 33%, in kelp forest coverage in recent years (Tangle, 2024). Kelp populations were not only affected by climatic shifts but also by declining coastal water quality in California. In the 1960s, sewage discharge entering coastal waters was one factor contributing to this deterioration (Foster & Schiel, 2010). A culmination of these incidents have negatively affected relationships that many organisms shared with the kelp forests, including the white abalone. Kelp forests are key for providing adequate habitat and food for any grazer-dependent and detritus-dependent trophic pathways (Graham, 2005). Abalone primarily consume drifting kelp, so to keep their population abundant, healthy, and dense, thriving kelp forests are required to supply a constant stream of essential nutrients (NOAA, 2025). When white abalone are in their juvenile stage, they are also very susceptible to predation. So, they tend to hide themselves in the holdfasts of giant kelp and bull kelp, as it is very difficult to get through as a larger predator. In order to have successful captive breeding programs, some focus needs to be applied to the ecosystem in which they are being placed. Environments most conducive to high biodiversity will mitigate direct threats to the white abalone.

While the spotlight is on kelp, it should be noted that large purple sea urchin populations are a direct threat to overall ecosystem health in kelp forests. Disproportionately large urchin populations will often decimate kelp forests by chewing off their holdfasts, leaving little to no food and protection for the abalone until regeneration occurs (Tasoff, 2020). The strategy to be implemented here is direct removal of urchins by commercial fisherman or hired professional divers. While divers are working on kelp or abalone restoration, they can also capture sea urchins if necessary. Reducing the purple sea urchin population will allow kelp forests to grow denser, creating more habitat for gastropods, mollusks, crustaceans, and fishes, etc., while also increasing the amount of drifting kelp available for abalone to feed on (Graham, 2005). Purple sea urchin, although smaller than red sea urchin, is still a delicate food and can also be used as fertilizer when ground up.

Range of Strategies

Strategy	Description
Planting Heat-Tolerant Kelp	Outplant heat-tolerant kelp species in degraded areas of the Santa Barbara Channel to restore kelp forest habitat and increase ecosystem resilience to warming ocean conditions.
Removal of Urchins	Reduce purple sea urchin populations through targeted removal to prevent overgrazing and allow kelp forests to recover.
Captive Breeding	Captive-bred abalone by working with existing programs like NOAA and the Cultured Abalone Farm. Outplant hatched juveniles once they reach approximately 25mm in size (Diskin, 2025).
Attachment Substrates	Introduce good crevice, low-relief, rocky terrain attachment substrates to encourage kelp settlement and growth, as well as protect juvenile abalone from predators.

Table 4 - Summary of Strategic Plans

No Action

Undertaking no preventative action will likely contribute to the continued population decline of the white abalone. Facing threats such as ocean warming, lack of kelp, and overpopulation of urchins, the white abalone is unlikely to improve in population with no action taken.

Planting Heat-Tolerant Kelp

Planting heat-tolerant kelp involves outplanting kelp strains that are more resilient to rising ocean temperatures and marine heat waves. As climate change increases the frequency and intensity of marine heat waves, many existing kelp forests experience large die-offs and slower recovery rates. Introducing kelp that can survive in warmer waters helps stabilize kelp forest ecosystems and maintain a consistent food supply for white abalone, which primarily feed on drifting kelp. By restoring kelp forest density and coverage, this strategy improves habitat quality for abalone and many other marine species while supporting overall biodiversity and ecosystem resilience. In order to select the best kelp for this strategy, we would utilize research findings and resources from partners such as the Jenn Caselle and Pisco Labs.

Removal of Urchins

The removal of purple sea urchins helps restore kelp forest ecosystems by reducing overgrazing pressure on kelp. In recent years, warming waters and declining predator populations have allowed purple urchin populations to grow rapidly, creating urchin barrens where kelp forests are heavily grazed and unable to recover. These conditions reduce available habitat and food for white abalone, which rely on drifting kelp. By removing urchins through diver-based removal efforts and supporting Indigenous harvesting practices, kelp forests can regrow, increasing kelp density and restoring ecosystem balance. These efforts and practices would be done in partnership with the Jenn Caselle and Pisco Lab divers, the Chumash Tribal Council, and Cultured Abalone Farm. This recovery improves habitat availability, increases food resources for abalone, and supports broader marine biodiversity.

Captive Breeding

White abalone captive breeding takes place in specialized laboratory facilities. It involves the introduction of a hydrogen peroxide solution to an individual, which then triggers sperm or egg release (White Abalone Captive Breeding Program, N.D.) Scientists determine which abalone to genetically cross using released sperm and eggs. Once fertilized, eggs are studied to determine healthy development. Outplanting of hatched juveniles occurs once they reach approximately 25mm in size (Diskin, 2025). Captive breeding is necessary to bring back white abalone populations.

Attachment Substrates:

Attachment substrates are natural or artificial surfaces designed to replace rocky or coral habitat terrain and encourage settlement of sessile species, while also offering protection for marine life (Bromley, 2006). The introduction of attachment substrates into habitats where abalone outplanting occurs is crucial to support kelp growth and reduce juvenile abalone mortality rates. Good crevice, low relief substrates would likely serve best for both abalone and kelp outplanting (M. Neuman, personal communication, March 5, 2026).

Strategy Selection

Our team used both a consequence table, *Table 5*, and a Multiple Criteria Decision Analysis, *Table 6*, to evaluate how each proposed strategy contributes to our fundamental objectives and overall conservation goals. The strategies we selected were chosen because we believe, when implemented cohesively, they have the potential to address the multiple factors contributing to the decline and endangerment of white abalone. However, even the partial implementation of these strategies could still play an important role in contributing to recovery efforts. Because if no action is taken, conditions and populations will continue to decline.

With the growing threat of climate change, if actions are not taken now, kelp forests will continue to degrade, marine heat waves will become more frequent, urchin barrens will expand

throughout the Santa Barbara Channel, and any existing wild white abalone populations will disappear, making captive breeding and long-term conservation efforts significantly more difficult. (Defranco, 2021).

Consequence Table

The consequence table, *Table 5*, evaluates how each proposed strategy performs relative to our fundamental objectives. Strategies are assessed qualitatively, using a scoring system of “+”, “-”, and “/”. A “+” indicates a strategy's positive contribution toward reaching said objective, a “-” indicates a negative contribution, and a “/” indicates little to no direct effect. By examining the positive, negative, and neutral outcomes, the table highlights which strategies address the greatest number of our fundamental objectives simultaneously.

Fundamental Objectives	Increase white abalone populations	Increase kelp forest cover	Reduce purple urchin populations	Restore natural reproduction rates	Control algal over-growth	Create resilient ecosystems that contribute to healthy and sustainable fishing/ fisheries	Increase eco-tourism	Restore Indigenous cultural traditions	Net Impact
Indicators	- Measure population density, abundance, and size.	-Percent forest cover, density per square mile.	- Population density of purple urchins in urchin barrels.	- Measure reproduction success through recruitment levels and larval monitoring.	- Algal % cover relative to giant kelp cover	- Stock assessments, bycatch and habitat impact assessments, & economic value of restored habitat	- Number of tourists annually.	- Indigenous Chumash levels of abalone use.	
Strategies									
No Action	-	-	/	-	-	-	/	/	-5
Planting heat-tolerant kelp	+	+	/	+	+	+	/	+	+6
Removal of purple urchins	+	+	+	-	/	+	/	+	+5
Captive Breeding	+	/	/	+	+	+	/	+	+5
Attachment substrates	+	+	/	+	/	+	/	/	+4

Table 5 - Consequence Table

As displayed in our consequence table, the “No Action” control strategy performs poorly across nearly all objectives. Displaying that, without intervention, white abalone populations will continue to suffer as they face the pressures of habitat loss, low population density, and ecological imbalance, all continually worsened by the growing threat of climate change.

Among our active management strategies, the outplanting of heat-tolerant kelp ranks highest, with the largest net-positive impact of (+6). This strategy targets the majority of our objectives, including increasing white abalone populations, increasing kelp forest cover, restoring abalone reproduction rates, controlling algal overgrowth, creating resilient ecosystems that contribute to healthy and sustainable fisheries, and restoring indigenous cultural traditions. While contributing far more directly to some of these objectives than others, for example, increasing kelp forest cover, this strategy can also indirectly target our conservation objectives. For example, as the climate warms, marine heat waves have become more common, wiping out existing kelp forests. By outplanting kelp species more tolerant to these warming waters, we protect abalones' primary food source and habitat, increasing their population, density, and natural reproduction rates.

Among our other strategies, the removal of purple urchins and implementation of captive breeding rank second, at a net impact of (+5). The removal of purple urchins is crucial, as it has the capacity to boost white abalone populations, increase kelp forest cover, manage urchin populations and barrens, and restore indigenous traditions, all while restoring a healthy ecosystem with healthy functions. Due to warming waters and the lack of predator populations, urchins have exploded in population, creating what are known as urchin barrens, where they kill vast areas of kelp and crowd out and starve abalone by outcompeting them. Through removal efforts conducted by marine science laboratories alongside Indigenous harvesting practices, this strategy can both support the revitalization of culturally significant food traditions and contribute to the restoration of healthy ecosystem functions.

Captive breeding has the most direct impact on restoring population density and reproduction rates of the white abalone. But while it has been the primary focus of many previous white abalone conservation plans, it's often prioritized to the extent that other ecological considerations receive less attention. For our conservation plan, we have stepped away from this idea, adopting a more holistic, ecosystem-based approach, where we will equally prioritize restoring and stabilizing the habitats where future captive breeding outplanting will occur. Within this framework, we have been considering how our plan can operate in unison with existing captive breeding programs, as we still recognize how crucial these efforts are, particularly for restoring reproduction rates, as white abalone are broadcast spawners that require high densities to breed. For these reasons, captive breeding is not classified as the strategy with the highest net positive impact, despite its direct benefits with abalone populations.

Lastly, introducing attachment substrates that support both kelp growth and serve as abalone habitat leaves us with a net impact of (+4). While this strategy seems to support our objectives less directly, introducing deep crevices and low relief substrates can be incredibly beneficial to kelp and abalone populations. These structures create stable surfaces where kelp can

securely attach, while providing crevices essential to juvenile abalone to hide and mature, especially since outplanting efforts release abalone recruits at such young ages. Overall, this strategy is an excellent choice for strengthening our overall habitat foundation, which is needed for long-term kelp forest recovery and successful abalone outplanting efforts with low mortality rates.

Multiple Criteria Decision Analysis

Similar to the consequence table, the Multi-Criteria Decision Analysis table, *Table 6*, is a tool we used to compare different strategies, based on how well they meet our set of defined objectives. However, unlike the consequence table, the MCDA quantitatively compares strategies and incorporates weighted scoring to reflect relative importance. This allows us to better analyze which strategies most effectively work toward our overall goals.

Multi-Criteria Decision Analysis (MCDA) for White Abalone

Fundamental Objectives	Weight (1-5)	Strategy Options (1-10)				
		No Action	Planting heat tolerant kelp	Removal of Urchins	Captive Breeding	Attachment substrates
Increase White Abalone populations	5	0	8 (40)	8 (40)	10(50)	7(35)
Increase kelp forest cover	4	0	10(40)	10 (40)	0(0)	10(40)
Reduce purple urchin populations	4	0	0	10 (40)	0(0)	0(0)
Restore natural reproduction rates (abalone)	3	0	4(12)	5 (15)	10(30)	8(24)
Create resilient ecosystems that contribute to healthy and sustainable fishing/ fisheries	3	0	8(24)	8 (24)	8(24)	5(15)
Control algal over-growth	2	0	3(6)	3 (6)	10(20)	0(0)
Increase eco-tourism	2	0	5(10)	0	4(8)	3(6)
Restore Indigenous cultural traditions	2	0	5(10)	0	8(16)	4(8)
Total		0	142	165	148	128

Table 6 - Multiple Criteria Decision Analysis (MCDA) for White Abalone

The MCDA table results generated totals from 0 to 165. As such, we score the range of strategies from least effective at 0 to most effective at 165. This wide range of effectiveness shows us that one strategy alone cannot complete all of the fundamental objectives. Employing multiple of the strategy options is needed to effectively address the diverse set of fundamental objectives. The objective given the highest weight is to increase White Abalone populations, with a weight of 5. Next, with weights of 4, are to increase kelp forest cover and reduce purple urchin populations. Less heavily weighted objectives are to restore natural reproduction rates of the White Abalone and to create resilient ecosystems that contribute to healthy and sustainable fishing/fishers, both earning 3's, and controlling algal over-growth, increasing eco-tourism, and restoring Indigenous cultural traditions, all with 2's.

The highest scoring strategy that most effectively addresses the fundamental objectives is the removal of urchins. This strategy obtained a score of 165 in the MCDA. Removing urchins strongly addresses the fundamental objectives of increasing kelp forest cover and reducing purple urchin populations, with scores of 10's in both, and also effectively addresses increasing White Abalone populations and creating resilient ecosystems, acquiring scores of 8's in both. By removing urchins by divers and introducing predation, this strategy will encourage healthy kelp regrowth by reducing pressure on them, balance predator-prey populations, and lead to stronger fisheries. Urchin removal will contribute both to kelp forest health and ultimately to the White Abalone population increase. Though the urchin removal strategy fails to address our objectives of increasing eco-tourism and restoring Indigenous cultural traditions, it scores consistently highly within other categories that are weighted higher.

With a score of 148, captive breeding is the second most effective strategy. Through our partnerships with NOAA and the Cultured Abalone Farm, which has experience in captive abalone breeding, our fundamental objective of increasing White Abalone populations is most directly addressed. This strategy also scored 10's in restoring abalone's natural reproduction rates and controlling algal overgrowth. As more abalones are introduced through captive breeding, algal overgrowth will decrease, as it is a food source for abalone. Captive breeding also effectively addresses our objectives of creating resilient ecosystems and restoring Indigenous cultural traditions, with scores of 8. Chumash traditions involve using abalone meat for food and shells for spiritual and artistic purposes (via VisitCalifornia), so an increased abalone population could allow for them to regain some of these traditions on a larger scale, as shells from deceased abalone would become more common. This strategy does not increase kelp forest cover or reduce purple urchin populations, two fundamental objectives that are key for the long-term well-being of the White Abalone, again showing that a wide variety of strategies must be implemented.

Planting heat-tolerant kelp was the third-highest-scoring strategy, at 142. This strategy directly relates to our fundamental objective of increasing kelp cover, scoring a 10. It would also,

in turn, create resilient ecosystems, as increasing the health of kelp would encourage a biodiverse, healthy ocean environment. However, this strategy would not address our objective of decreasing purple urchin populations.

The lowest-scoring strategy (other than “no action”) was introducing attachment substrates, which scored 128. Though attachment substrates would aid in kelp forest cover (scoring a 10), as they provide solid anchors for kelp to grow on, the strategy does little to directly address other categories, such as reducing purple urchins and controlling algal overgrowth.

The “no action” strategy resulted in a 0 score, making it inconsequential for the plan’s goals. This MCDA result concludes that taking no action is not a feasible conservation strategy.

Data and Knowledge

Our information and data come from a variety of sources, including academic journals, scientific reports, and government agencies like NOAA and the California Department of Fish and Wildlife. The majority of our sources were found through the University of California, Santa Barbara Library database. Our search processes included first broad term searches for white abalone and the Santa Barbara Channel, and later narrower searches based on our findings. These narrow searches included diving deeper into the relationships among kelp forest health, warming waters, sea urchin populations, breeding strategies, and disease, and how these factors affect abalone populations and captive breeding efforts.

We also had the opportunity to interview Dr. Melissa Neuman, an Endangered Species Recovery Coordinator at NOAA, with the “primary responsibility of implement[ing] NOAA’s strategy for conserving and restoring endangered white abalone, an iconic member of kelp forest communities in the U.S. and Mexico, through captive propagation and enhancement activities.” (NOAA, n.d.). Dr. Neuman provided us with a plethora of information, knowledge, and firsthand experience with white abalone populations, which we used to both guide and support our existing conservation plan. She informed us of NOAA’s existing captive breeding programs, strategies for releasing abalone recruits, qualitative protocols used to determine outplanting sites, and monitoring techniques used to measure population density and mortality rates, including shell collection and use of EDNA. Dr. Neuman was also instrumental in helping us strengthen the connection between kelp forest conservation and our white abalone conservation plan, adding an additional element and ecological component that former plans have lacked. (M. Neuman, personal communication, March 5, 2026).

Risks and Uncertainties

There are several risks and uncertainties associated with the conservation of white abalone that are outside of the control of the conservation plan. These risks and uncertainties could be a cause for failed efforts to increase the population back to healthy levels. The threats vary in severity and should be addressed in accordance with this. The risks are marine heat waves, disease outbreaks, warming oceans, lack of data, and knowledge.

Risk 1: Marine Heat Waves (Severe)

Marine heat waves pose a serious threat to the success of our kelp restoration plan. There is an increase in the frequency, intensity, and duration of marine heat waves in the last four decades. Both surface and bottom marine heat waves occur in the Santa Barbara Channel. Bottom marine heat waves cause direct thermal stress, which can result in mass mortality of Giant kelp. In addition, areas that experience a higher frequency of marine heat waves typically have slower recovery rates. These events are known to cause numerous negative impacts, resulting in the mass degradation of kelp. Areas that have experienced frequent marine heat waves show slower kelp forest recovery rates. Kelp forests serve as an essential habitat for the white abalone, which is why a loss of kelp habitat would severely reduce the success of restoration.

Risk 2: Disease outbreak/withering syndrome and sea star wasting syndrome(severe)

Disease outbreak is a major threat to white abalone recovery. One of the most impactful diseases is withering syndrome, historically causing significant population decline. Another disease that poses a threat more indirectly to white abalone is sea star wasting disease. This disease has decimated the Sunflower Sea Star(*Pycnopodia helianthoides*) populations, leading to the overpopulation of purple urchins. The overpopulation of purple urchins leads to the overgrazing and degradation of kelp forests, leading to the formation of an urchin barren where kelp is unable to recover.

Risk 3: Low survival rate among released captive-bred white abalone (moderate)

Since white abalone fail to reproduce successfully naturally, captive breeding is an essential part of this restoration process. A risk associated with captive breeding is the possibility of low survival rates among the released captive-bred white abalone. They are released at around 1 year old or 25mm in size. Since they are being released at a younger age, they are more susceptible to predation, environmental stress, or failure to adapt to the environment. If survival rates remain low, restoration sites could fail to establish self-sustaining populations. One strategy to overcome this is to fabricate shelters in which the adolescent white abalone can seek refuge away from predators.

Risk 4: Lack of Data and Knowledge (Moderate)

Since there are such a low number of white abalone in the wild, individuals are rare and difficult to locate. Their population distribution, reproductive success, and long-term survival are difficult to track. This causes uncertainty in making restoration decisions without complete data or to evaluate whether restoration efforts are successful. Kelp forests are highly dynamic and influenced by many environmental factors which can cause difficulty in predicting how white abalone will respond to unstable environmental conditions. In order to reduce uncertainty, regular monitoring and research is essential. Partnerships through NOAA can help to expand datasets of abalone populations as well as implementing adaptive management strategies, adjusting restoration plans based on current scientific findings.

Conclusions/Future Steps

Despite challenges the white abalone faces in restoring its population back to healthy numbers, the community, NOAA, and other partners in the white abalone breeding program are determined to restore this ecosystem engineer. The Fundamental objectives in this plan are pathways to increase its population through breeding, habitat restoration, and monitoring to create a resilient ecosystem. The planting of heat-tolerant kelp, urchin removal, and breeding programs all work together to support a healthy population that will positively impact human well-being through restoring indigenous traditions, fisheries, and ecotourism. By incorporating local shareholders and working with government agencies like NOAA, we can achieve our fundamental goals and bring back wild white abalone populations to the Santa Barbara Channel.

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